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A Knowledge-Guided refinement framework for Late-Season Rice–Weed segmentation using UAV imagery under canopy closure

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ABSTRACT

Accurate late-season rice–weed mapping is particularly important for identifying residual weed escapes that directly influence targeted herbicide application and final yield outcomes, enabling site-specific management and reducing unnecessary chemical inputs. While deep learning approaches perform well at early growth stages, they frequently generate unstable predictions and overestimate residual weeds under late-season conditions. This study presents an operational, knowledge-guided refinement framework for late-season rice–weed segmentation from UAV RGB orthomosaics. A compact U-Net is used to produce an initial semantic prediction, which is subsequently refined using interpretable agronomic and photographic priors. The refinement stage applies vegetation gating with a dark-pixel safeguard to suppress water and non-vegetated surfaces, followed by the integration of soft chromatic cues and crop-row-informed geometric constraints to stabilize class boundaries while preserving plausible inter-row weed occurrences. The framework is evaluated using a late-season UAV orthomosaic and co-registered reference labels from an irrigated rice paddy in Anqing, China, tiled into leakage-safe training and validation subsets with three classes (rice, weed, and other). On independent test tiles, the refined outputs improve spatial consistency relative to appearance-only baselines, achieving an overall accuracy of 0.974, a macro-F1 score of 0.937, and a mean intersection-over-union of 0.883. The workflow operates directly on georeferenced UAV imagery and produces GIS-ready classification and probability layers in GeoTIFF format, supporting large-area mapping and integration into agricultural decision-support systems. Scripts for the refinement stage are released to facilitate reproducibility and comparison and are available at https://github.com/mnasarahmad/WeedGraphaNet; we also outline extensions toward multi-site validation and learned priors for glare and shadow handling.

1. Introduction

Weeds remain a persistent constraint on rice productivity and a major source of avoidable herbicide loading in irrigated landscapes worldwide (Kraehmer et al., 2016; Pervaiz et al., 2024). Repeated studies in paddy systems show that herbicide residues from paddy fields can move into canals and other waters through drainage and storm events, elevating ecological risks and increasing water treatment costs for downstream users (Voccia et al., 2024). This motivates site-specific weed management (SSWM) that confines inputs to where they are needed rather than relying on blanket sprays (Huang et al., 2018). Evidence from agronomy and weed science demonstrates that spatially selective approaches can lower chemical use and still sustain efficacy when reliable maps guide decisions for weed control. The environmental impetus is therefore clear: accurate delineation of weed patches at operational scales can reduce the footprint of interventions, limit off-field transport, and support safer water in rice-dominated basins (Sudo et al., 2018). Table 1 summarizes the UAV weed-mapping studies in recent years.

Over the last decade Table 1, UAV imaging paired with deep learning has delivered strong early-season weed mapping results in rice when targets are exposed and separability from background is high (Stroppiana et al., 2018). Lightweight U-Net variants and related segmenters operating on RGB or compact multispectral payloads have achieved high accuracy and even near real-time inference in open canopies, establishing a robust capability baseline for paddy systems (Lan et al., 2021). At the same time, systematic reviews of UAV-based weed detection confirm that much of the literature concentrates on early

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vegetative stages and visually easy scenes, which simplifies generalization and evaluation but underrepresents later phenology (Huang et al., 2020; Yu et al., 2022). Consequently, late-season conditions remain weakly supported, despite being the phase in which surviving escapes most strongly influence intervention choices.

Late-season conditions are fundamentally harder: the rice canopy closes, weeds are interspersed or occluded, and flooded surfaces introduce optical artifacts that destabilize RGB inferences (Li et al., 2024b; Xing, 2024. Specular sun-glint from water is a first-order confounder in near-water mapping and has been repeatedly documented in UAV and satellite studies, with recent work proposing explicit glint detection or correction for low-altitude imagery (Ortega-Terol et al., 2017). In addition, exposure/white-balance swings, cast shadows, and motion softness vary across sorties and suppress the already weak class margins that deep models rely on (Duan et al., 2020; Genze et al., 2022). Without explicit treatment, these acquisition-condition effects lead to unstable segmentations and over- or under-estimation of residual weeds precisely when operations consider targeted treatments or drone-based prescriptions for escapes. Consequently, late-season mapping demands approaches that can both recognize weak appearance cues and discount acquisition artifacts that mimic vegetation.

A complementary, underused signal at canopy closure is agronomic structure. Rice fields exhibit row geometry that is spatially coherent and persistent across dates, while late-season weeds frequently appear between rows, along bunds or inlets, and as features with characteristic size–shape–adjacency patterns (He et al., 2025; Zhang et al., 2021). Research communities have made notable progress on robust crop-row detection under variable lighting, curvature, and partial occlusion, suggesting that row cues can stabilize downstream perception tasks when pixel-level separability collapses (Louargant et al., 2018; Shi et al., 2023). Related work in remote sensing also shows that injecting prior knowledge (geometric, physical, or task-level) into segmentation networks improves reliability under nuisance conditions and limited contrast (Cui et al., 2021; Dyson et al., 2019). This reflects a broader methodological shift from traditional appearance-only segmentation toward reasoning-based approaches that integrate row alignment, shape, texture, and greenness cues, which are particularly effective under late-season rice canopy conditions (Kraehmer et al., 2016).

Late-season rice–weed segmentation is affected by several characteristic failure mechanisms, including (i) false positives over non-vegetated surfaces caused by water reflections and illumination variability, (ii) reduced spectral separability between rice and weeds under canopy closure, and (iii) spatial inconsistency due to occlusion and weak class boundaries. To address these challenges, this study focuses on integrating interpretable cues that directly target these failure modes. Specifically, vegetation gating suppresses non-vegetated artifacts, chromatic cues help differentiate subtle green variations between rice and weeds, and geometric constraints based on crop-row structure improve spatial consistency and plausibility of predictions.

Finally, we evaluate the proposed framework on late-season paddies using per-class metrics (rice/weed/others) and an operational indicator: reduction in spray-eligible area relative to a strong U-Net baseline. This connects model performance to environmental relevance by quantifying the potential to shrink treatment footprints and thereby lower herbicide loading and off-field transport. Our framing aligns with ongoing advances in site-specific treatment of late-season escapes, including drone-based applications, and with broader agronomic evidence that selective control can reduce chemical inputs without sacrificing efficacy. By design, the method is RGB-only and interpretable, which facilitates transfer to smallholder and mixed-farming contexts where heavier sensors or opaque pipelines are difficult to sustain.

The specific objectives of this study are:

Design a knowledge-guided RGB-only segmentation framework that integrates condition-aware cues (e.g., glare, exposure, shadow) and agronomic constraints (e.g., row structure and component geometry) for robust late-season canopy-closure mapping.

Compare multiple model configurations, including appearance-only U-Net baselines and the proposed knowledge-guided framework, using OA, Micro/Macro-F1, mIoU, and per-class Precision/Recall/F1/IoU, with ablation analysis to quantify the contribution of each component.

Table 1
Prior UAV weed-mapping studies, key gaps at canopy closure, how this work addresses them, and reported metrics from the original publications. Reported values are included for qualitative reference only and are not directly comparable because datasets, crops, sensors, class definitions, and evaluation protocols differ.

Reference	Study description	**Reported metric (s)**	Identified gaps	Current study contribution
(Shahi et al., 2023)	Compared multiple deep learning models (e.g., U-Net, DeepLabV3 +) for UAV-based weed segmentation using RGB imagery.	mIoU = 56.21%, F1 = 88.24%	High-performing models still limited by moderate mIoU (56.21%); sensitive to data quality and field artifacts; no explicit handling of illumination or agronomic structure.	Introduces photo priors (glare/shadow masks) and agronomic reasoning to stabilize late-season segmentation and improve interpretability.
(Nuankaew et al., 2025)	Developed and compared deep learning models (DenseNet121, InceptionV3) for rice–weed classification in aerial images.	Accuracy = 97.94%	Focused on accuracy, but not field artifacts (glint, canopy closure) or interpretability; lacks integration of field priors.	Integrates inexpensive, interpretable knowledge sources (photo priors, agronomic rules) to address artifacts and support robust, site-specific mapping.
(Genze et al., 2022)	Evaluated deep learning weed segmentation under motion blur and occlusion in sorghum fields.	F1 > 89%	Models misclassify plant borders; no explicit row/edge reasoning; no agronomic priors.	Applies component-level plausibility constraints (row orientation, adjacency, size) to recover subtle escapes and suppress implausible predictions.
(Bah et al., 2018)	Unsupervised CNN training for weed detection using UAV images with automatic crop-row detection.	AUC-based results reported; not directly comparable	Emphasizes inter-row weeds; limited intra-row handling and late-season occlusion; no photo-based priors.	Models both scene photography (photo priors) and crop structure (agronomic reasoning) to tackle intra-row and late-season challenges.
(Murad et al., 2023)	Systematic review of DL for weed detection; highlights rapid progress with RGB imagery.	Not directly reported (review paper)	Appearance-only focus; sensitive to lighting/background; limited interpretability and transferability; few use explicit knowledge infusion.	Proposes a knowledge-infused, RGB-only, interpretable framework that explicitly models scene and crop structure for better transfer.
(Gopalakrishnan et al., 2025)	Developed AWRC-DLMLO: a deep learning model using RA-UNet, ShuffleNetV2, lemurs optimization, and cascading Q-network for weed/crop classification.	OA = 99.11%, F1 = 97.33%	Higher complexity; lacks interpretability and field-specific priors.	Keeps model simple and interpretable via deterministic refinement and agronomic priors suitable for constrained deployment.

Demonstrate operational relevance by quantifying the reduction in spray-eligible area relative to the best baseline and providing reproducible artifacts (trained model and refinement scripts).

2. Materials and data

2.1. Study area

The field campaign took place at Jiu Farm, Anqing, Anhui, China (humid-subtropical monsoon; irrigated, low-relief alluvial plain). Fields are leveled and bunded, supplied by lined canals with controlled inlets/outlets; late-season water depth is shallow and spatially variable along furrows and edges. Rice was machine-transplanted in early May 2025 (local practice). On the survey date, the crop was 2 months 25 days old (late season; canopy closure), when residual weeds persist mainly between rows and along bunds/field margins. Typical nuisance backgrounds include shallow water, wet soil, stubble/boards, and compacted bund tops.

2.2. UAV imagery acquisition and preprocessing

UAV and ground imagery were acquired on 25 July 2025 between 09:30–13:00 local time using a DJI Mavic 3 Multispectral. Although part of the acquisition window is close to solar noon, moderate shadow variability remains across the scene. To mitigate its impact, the proposed framework incorporates exposure/shadow reliability cues and a dark-pixel safeguard within the vegetation gating stage, reducing false vegetation responses in shaded regions.

To maintain an RGB-only pipeline, only the RGB camera (20 MP, 5472 $\times$ 3648 px; 4/3″ CMOS) was used. Flights at 30 m AGL with 80% forward and 70% side overlap yielded a ground sampling distance (GSD) of approximately 1.5 cm. Orthomosaics were generated in Agisoft Metashape v1.8 (SfM/MVS), georeferenced with ground control points (GCPs), and exported as GeoTIFFs. Panel images acquired at take-off supported simple white-balance and reflectance normalization.

From the full orthomosaic, a single analysis patch (red-rectangle outline in Fig. 1 emphasizes the field-scale context and the spatial extent of the analysis area used for segmentation.) was clipped, resampled to 1.5 cm, and tiled into fixed-size patches (512 $\times$ 512 px) for modeling. A spatially blocked 80/20 train/validation split was used to reduce spatial

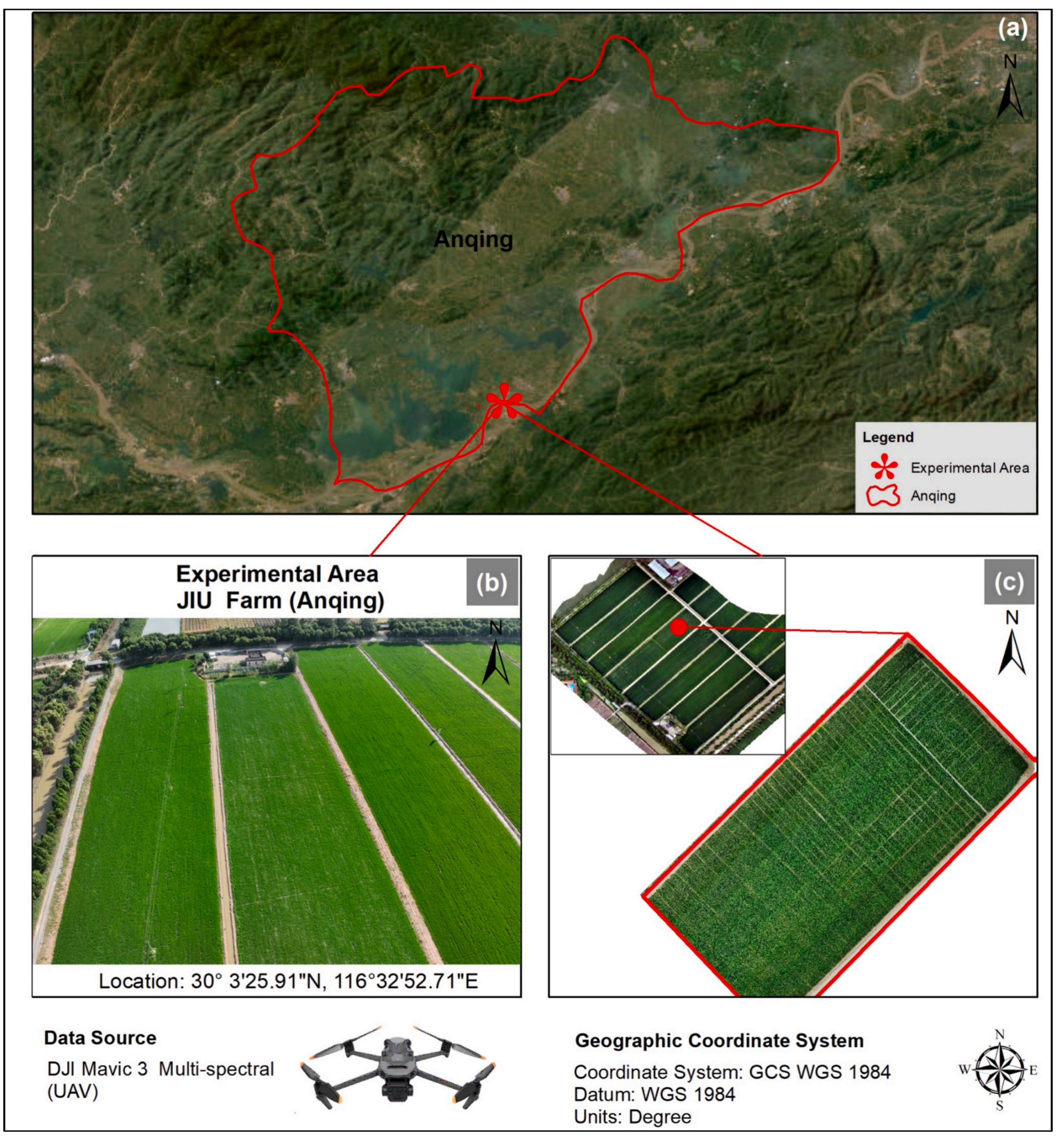

Fig. 1. (a) Location of JIU Farm Anqing, China; (b) oblique view of the experimental field; and (c) selected analysis patch (outlined in red) used for UAV-based segmentation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

leakage, and an additional cross-tile split was prepared for robustness checks.

2.3. Ground images for seeded cues

On the same date, ground-level camera images documented the in-situ appearance of weed, rice, and “others” (water, soil/bunds, boards/stubble) under prevailing illumination and water conditions (Fig. 2). From these we derived three lightweight cues later used in WeedGraphNet’s reasoning stage (Sec. 3): (i) glare masks over specular water highlights, (ii) exposure/shadow reliability maps, and (iii) class-appearance hints (color/texture prototypes). These cues act as soft weights/gates to stabilize decisions where RGB appearance is ambiguous at canopy closure.

2.4. Ground reference and labels

Contemporaneous field inspection and high-zoom visual interpretation were used to digitize labels as polygons in ArcGIS Pro for three classes: Weed, Rice (late-season canopy, machine-transplanted, 2 months 25 days), and Others (water, soil/bunds, boards/stubble). The annotated polygons comprised approximately 220 instances (Rice: ~60, Weed: ~109, Others: ~51). Polygons were rasterized at 1.5 cm GSD, co-registered to the raster patch, and quality-checked (local affine adjustments as needed).

The final labeled set contains 1,384,278 labeled pixels, with class proportions of 90.88% Rice, 4.14% Weed, and 4.98% Others (Table 2). These correspond to mapped areas of 283.0 m^2 (0.0283 ha) for Rice, 12.9 m^2 (0.0013 ha) for Weed, and 15.5 m^2 (0.0016 ha) for Others. Connected-object counts (8-connected) in the rasterized labels are consistent with the original polygon instances and total approximately 220. The dataset was partitioned into 400 training and 20 validation tiles (420 tiles in total). Here, the value “220” refers to connected object instances, not the number of tiles.

2.5. Derived structural and condition layers

To support the seeded reasoning module, we computed: (i) row-alignment coherence (dominant orientation and periodicity) from structure-tensor/Sobel gradients; (ii) local texture entropy (sliding window) to distinguish uniform crop rows from heterogeneous patches; and (iii) glare/exposure/blur maps from the UAV imagery, cross-checked against the 25 July 2025 ground images. Other agricultural reasonings to differentiate between classes were added as: for instance, any pixel that does not belong to vegetation (weed or rice) should refer to others. All ancillary layers, where applicable, were co-registered to the 1.5 cm base grid using nearest-neighbor resampling.

2.6. Dataset assembly and tiling protocol

We constructed learning samples from the labeled patch using a fixed tiling protocol and a leakage-safe split. Raster tiles were exported at 512 $\times$ 512 px with no overlap (stride = 512). Each tile consists of an RGB image (8-bit GeoTIFF) and a single-band label (PNG) with the encoding {0 = unlabeled, 1 = Weed, 2 = Rice, 3=Others}. Edge tiles were reflect-padded to preserve spatial size; label padding replicated edge values to avoid artificial boundaries.

To avoid trivial tiles and reduce class sparsity, a manifest builder retained tiles according to per-tile class presence and quality checks:

1. Class presence filter. Tiles were kept if at least one target class occupied $\geq$ 1% of pixels; fully unlabeled tiles were discarded.
2. Label validity. Any out-of-range label codes were remapped to 0 (unlabeled) and such tiles were excluded.
3. Consistency. Image–label dimensions were verified prior to export.

The final dataset comprises 400 training and 20 validation tiles (**Section 2.4**), created using a blocked spatial split to minimize spatial leakage and ensure geographic independence between sets. Although the number of tiles is modest, each tile contains dense pixel-level annotations at 1.5 cm resolution, resulting in over 1.38 million labeled pixels. This supports a reliable evaluation despite the limited number of tiles.

In addition, the use of a compact network architecture, data augmentation, and regularization helps mitigate overfitting under this dataset size. File paths are listed in plain-text manifests, enabling reproducible data loading. This assembly preserves the natural class imbalance (rice-dominant scenes) while ensuring that Weed and Others are present in both sets for fair evaluation.

Fig. 2. Ground images to derive cues for reasoning in WeedGraphNet.

Table 2
Ground-truth label summary at 1.5 cm GSD: per-class pixel counts, area, and 8-connected object instances (derived from rasterized polygons). The dataset comprises 420 tiles (400 training and 20 validation).

Class	Pixel count	Share of labelled pixels (%)	Area (m^2)	Area (ha)	Connected objects
Weed	57,324	4.14	12.9	0.0013	109
Rice	1,257,970	90.88	283.0	0.0283	60
Others	68,984	4.98	15.5	0.0016	51
Total (labeled)	1,384,278	100.00	311.4	0.0312	220

3. Methodology

This study introduces WeedGraphNet, a knowledge-infused segmentation pipeline designed for late-season rice–weed mapping under canopy closure using high-resolution UAV RGB orthomosaics. The framework combines a compact U-Net with a lightweight refinement stage that incorporates vegetation gating, chromatic priors, and agronomic geometric constraints to improve segmentation robustness under challenging field conditions. Rather than relying solely on appearance-driven deep learning, the proposed method combines a compact U-Net with a reasoning-based refinement stage that explicitly models agronomic structure and acquisition conditions. This design directly targets failure modes common in late-season scenes, including canopy occlusion, shallow flooded surfaces, and illumination-related artifacts.

Fig. 3 conceptually illustrates the methodological shift from traditional appearance-only segmentation toward the proposed reasoning-based segmentation paradigm. In this paradigm, pixel-level predictions are complemented by higher-level cues such as row alignment, geometric structure, texture, and greenness that remain informative when spectral separability deteriorates under canopy closure. This conceptual framing motivates the integration of interpretable agronomic and photographic priors into the segmentation process.

Building on this concept, Fig. 4 presents the complete WeedGraphNet framework, which couples a U-Net appearance model with a lightweight refinement stage informed by domain knowledge. The framework incorporates vegetation gating, chromatic cues (green versus yellow-green), spatial context related to crop rows and field edges, and simple morphological constraints to stabilize rice–weed boundaries while suppressing implausible predictions. Input data consist of a single georeferenced UAV orthomosaic and a co-registered label raster (codes: 0 = unlabeled, 1 = Weed, 2 = Rice, 3 = Others), which are tiled into leakage-safe training and validation sets to support reproducible learning and evaluation.

3.1. Data assembly and tiling

The orthomosaic and label raster were clipped to the analysis patch and exported to non-overlapping tiles of 512 × 512 px (stride = 512) to standardize network input and enable memory-safe training/inference. A blocked spatial split was used to minimize leakage: tiles located within the same local neighborhood were kept in the same partition. Tiles were retained if at least one target class occupied ≥ 1% of pixels; unlabeled tiles and tiles with invalid codes were discarded. Absolute file paths were stored in plain-text manifests (train/val) for exact reproducibility. The natural class imbalance (rice-dominant scenes) was preserved intentionally to reflect operational conditions at canopy closure.

3.2. Data appearance model (U-Net)

The segmentation backbone is a four-level U-Net (encoder–decoder with skip connections; 3 × 3 convolutions with ReLU) producing a three-way softmax (weed/rice/others). Tiles of 512 × 512 × 3 RGB are used with an encoder depth of 4 (64–128–256–512 filters), "same" padding, and skip concatenations. Training used Adam (initial learning rate 3×10^{-4}), batch size = 8, class-weighted cross-entropy to mitigate imbalance, L2 = 2×10^{-4}, and early stopping (patience = 8) with best-checkpoint restore. Augmentation comprised small rotations, flips, translations, and mild brightness/contrast jitter to improve robustness without erasing chromatic differences that help separate rice (deeper green) from weeds (yellow-green). Short schedules (30–60 epochs) were sufficient for convergence. For class $k \in \{1,2,3\}$, the softmax output is:

$$p_k(x) = \frac{\exp(z_k(x))}{\sum_{j=1}^{3}\exp(z_j(x))} \quad (1)$$

The learning objective is the class-weighted cross-entropy:

$$\mathscr{L}_{\mathscr{CE}} = -\frac{1}{|X|}\sum_{x\in X}\sum_{k=1}^{3} w_k \, 1[y(x) = k]\log p_k(x) \quad (2)$$

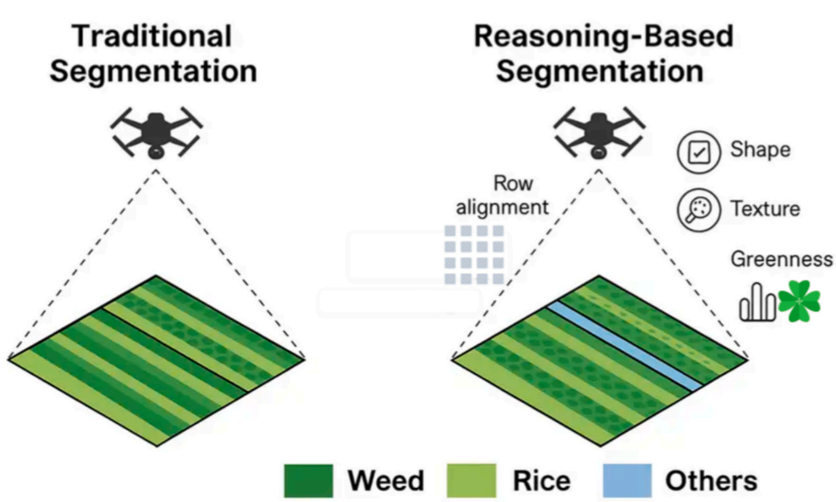

Fig. 3. Traditional (left) relies on greenness/texture and falters under canopy closure; WeedGraphNet (right) adds row alignment and shape/texture cues, yielding cleaner and high accuracy rice–weed–other segmentation.

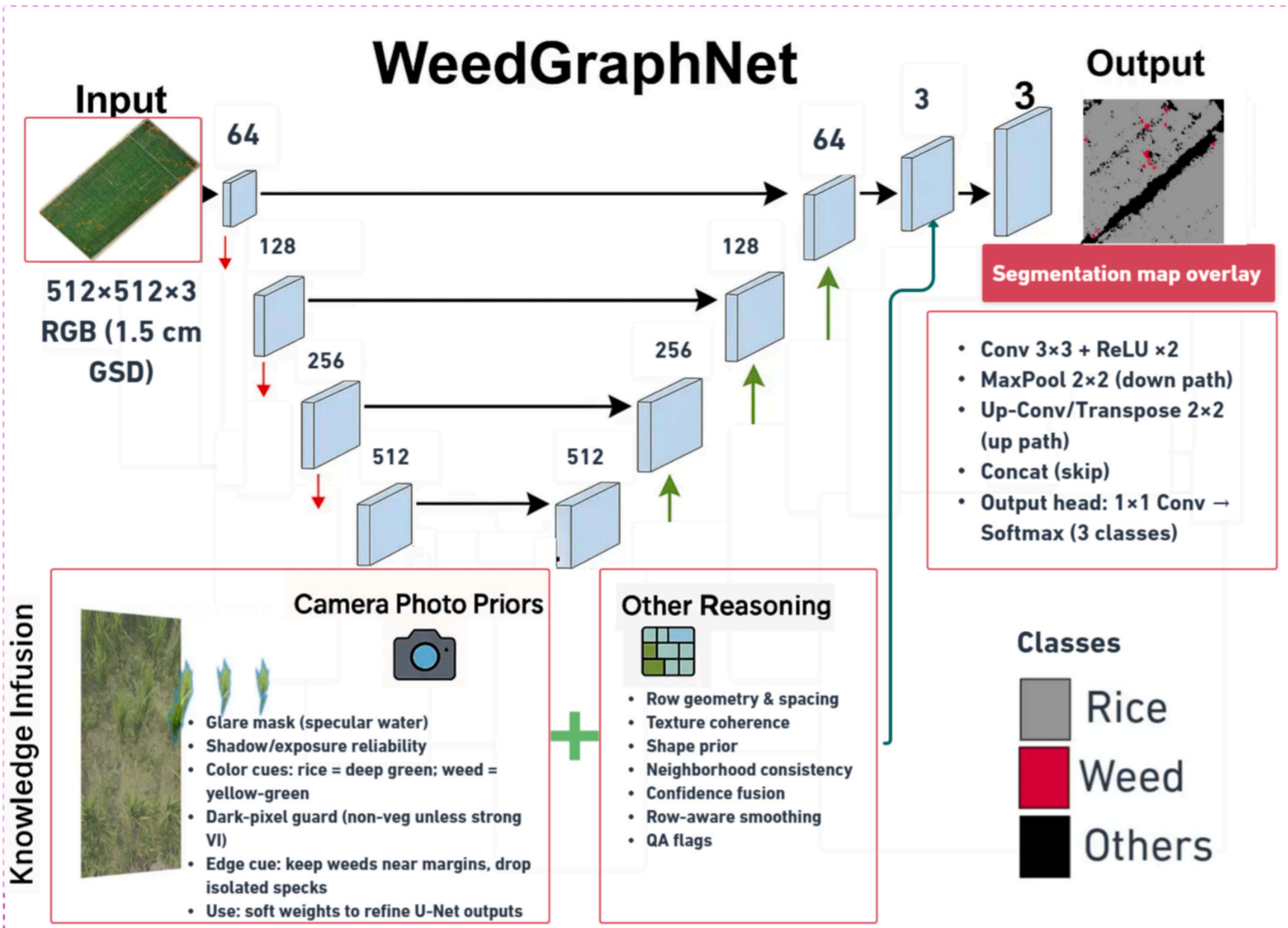

Fig. 4. WeedGraphNet framework: a 4-level U-Net on 512 × 512 RGB with knowledge infusion (camera-photo priors + agronomic reasoning) to produce the final Rice/Weed/Others segmentation map.

where X denotes the set of pixels in a training tile or mini-batch, x is a pixel location, $y(x)$ is the ground truth class label at pixel x , and w_k is the class weight for class k. The indicator function $1[y(x) = k]$ equals 1 if the true label at x is class k, and 0 otherwise. The predicted probability for class k is denoted by $p_k(x)$, and $z_k(x)$ represents the corresponding logit:

$$w_k = \frac{1/\mathrm{freq}(k)}{\sum_{j=1}^{3} 1 \Big/ \mathrm{freq}(j)} \tag{3}$$

Where freq(k) denotes the frequency of class k in the training data. We optimize the L2-regularized objective:

$$\mathscr{L} = \mathscr{L}_{\mathscr{C}\mathscr{E}} + \lambda\|\theta\|_2^2 \tag{4}$$

where λ is the regularization coefficient and $\|\theta\|_2^2$ denotes the L2 norm of model parameters θ.

3.3. Knowledge-infused refinement

To stabilize predictions in scenes with occlusion, shallow water, shadows, and weak class margins, the softmax map is refined component-wise with interpretable rules:

4. Vegetation gate. A vegetation index is used to suppress low-vegetation pixels classified as Others. In this study, vegetation gating relies on RGB-derived indices (e.g., VARI and ExG). We compute VARI as:

$$\mathrm{VARI} = \frac{\mathrm{G} - \mathrm{R}}{\mathrm{G} + \mathrm{R} - \mathrm{B}} \tag{5}$$

We also use Excess Green (ExG):

$$\mathrm{ExG} = 2\mathrm{G} - \mathrm{R} - \mathrm{B} \tag{6}$$

where R, G and B denote the red, green, and blue channel values, respectively.

The gate with a dark-pixel guard is:

$$Vegetation(x) = \begin{cases} 1 if VI(x) > T_{vi} and Y(x) > T_{Dark} \\ 0, otherwise \end{cases} \tag{7}$$

with luminance Y computed as $Y = 0.2126R + 0.7152G + 0.0722B$.

where $VI(x)$ denotes the vegetation index (e.g., VARI or ExG), $Y(x)$ is the luminance value at pixel x and T_{vi} and T_{dark} are threshold values for vegetation detection and dark-pixel suppression, respectively.

5. Chromatic prior (soft). Within VI-positive pixels, deeper/cleaner green ($G \gg R$, $G \geq B$, high $(G - R)/(G + R)$) nudges toward Rice; yellow-green (higher R/G) nudges toward Weed. These biases never override strong network confidence.

6. In this study, vegetation gating relies solely on RGB-derived indices (e.g., VARI and ExG).

7. Spatial ecology & morphology. Small Weed components touching rice boundaries are kept (edge escapes); isolated specks far from vegetation are demoted. A min-area filter (≤5–8 px) removes tiny

components. Size/shape/entropy cues prefer compact, row-coherent Rice over speckled Weed. Long gray strips and hard surfaces (levees/paths) are constrained to Others by combining low-VI masks with continuity checks.

Thresholds are modest and tile-adaptive (e.g., luminance percentiles for dark-pixel guards). The refinement acts as a stabilizer, not a second classifier.

3.4. Inference and products

Inference uses a fixed-size window of 512 px with reflect padding at tile boundaries, enabling memory-safe deployment on large orthomosaics. The workflow produces (i) per-class probability rasters and (ii) a refined single-band thematic map with the original {1,2,3} encoding, all exported as georeferenced GeoTIFF layers aligned to the coordinate reference system of the input orthomosaic. These outputs support direct integration into GIS environments for area-based summaries, weed-cover mapping, and generation of prescription layers. All processing steps are reproducible using the provided data manifests and configuration files.

3.5. Baselines and ablations

To evaluate the contribution of the proposed knowledge-guided refinement, we include an appearance-only baseline and an inference-time ablation study while holding the trained U-Net backbone fixed. The baseline (A0) corresponds to the U-Net output without refinement. Refinement modules are enabled/disabled at inference to form: (A1) vegetation gating (VG) only, (A2) VG + chromatic prior (CP), (A3) VG + geometric/component constraints (GC), and (A4) full WeedGraphNet (VG + CP + GC). Two baselines are considered in this study: (i) the appearance-only U-Net output, and (ii) U-Net followed by generic morphological cleaning, which serves as a standard post-processing comparator to distinguish learning-based improvements from generic post-processing effects. The morphology-based baseline consists of size filtering and simple morphological operations applied to the U-Net output without VG, CP, or GC. All variants are evaluated on the same leakage-safe held-out tiles using OA, Micro-F1, Macro-F1, mIoU, and Weed precision/IoU.

3.6. Accuracy assessment

We evaluate segmentation on the held-out validation tiles using pixel-level Overall Accuracy (OA), Micro-/Macro-F1, mean Intersection-over-Union (mIoU), and per-class Precision, Recall, F1, and IoU. Let $K = 3$ with classes $k \in \{\text{Weed}, \text{Rice}, \text{Others}\}$, and let TP_k, FP_k, FN_k, TN_k denote confusion-matrix counts for class k. The per-class metrics are:

$$\mathrm{Precision(k)} = \frac{\mathrm{TP_k}}{\mathrm{TP_k} + \mathrm{FP_k}} \tag{8a}$$

$$\mathrm{Recall(k)} = \frac{\mathrm{TP_k}}{\mathrm{TP_k} + \mathrm{FN_k}} \tag{8b}$$

$$\mathrm{F1(k)} = \frac{2\,\mathrm{Precision(k)}\,\mathrm{Recall(k)}}{\mathrm{Precision(k)} + \mathrm{Recall(k)}} \tag{8c}$$

$$\mathrm{IoU(k)} = \frac{\mathrm{TP_k}}{\mathrm{TP_k} + \mathrm{FP_k} + \mathrm{FN_k}} \tag{8d}$$

Overall Accuracy (OA) is computed over all pixels x(total N) as

$$\mathrm{OA} = \frac{1}{N}\sum\nolimits_{x=1}^{N} 1\left[\hat{y}(x) = y(x)\right] \tag{9}$$

Micro-averaged scores first aggregate counts across classes,

$$TP_\bullet = \sum\nolimits_{k=1}^{K} TP_k, \quad FP_\bullet = \sum\nolimits_{k=1}^{K} FP_k, \quad FN_\bullet = \sum\nolimits_{k=1}^{K} FN_k \tag{10a}$$

then compute

$$\mathrm{F1_{micro}} = \frac{2\,TP_\bullet}{2\,TP_\bullet + FP_\bullet + FN_\bullet} \tag{10b}$$

Macro-averaged F1 and mean IoU average per-class values:

$$\mathrm{F1_{macro}} = \frac{1}{K}\sum\nolimits_{k=1}^{K} \mathrm{r}F1(k) \tag{10c}$$

$$\mathrm{mIoU} = \frac{1}{K}\sum\nolimits_{k=1}^{K} \mathrm{r}IoU(k) \tag{10d}$$

where r denotes the class index corresponding to the segmentation classes (Weed, Rice, Others). Both count and row-normalized confusion matrices are reported to expose failure modes among Weed, Rice, and Others. For qualitative diagnostics, class-wise probability maps and error overlays (false positives/negatives) are included alongside summary metrics.

4. Results

4.1. Segmentation performance of WeedGraphNet

This section presents the quantitative performance of the proposed framework, supported by corresponding analysis in Tables 3–6 and Fig. 5. The quantitative performance of the proposed framework achieved OA = 0.9742, Micro-F1 = 0.9742, Macro-F1 = 0.9368, and mIoU = 0.8833 on the held-out validation tiles (Table 3). These values indicate consistent agreement between predictions and labels across classes despite the strong class imbalance (rice-dominant scenes at canopy closure). The high OA aligns with prior studies that U-Net-type encoders trained on high-resolution UAV imagery can reach > 0.9 accuracy for crop/weed segmentation when tiling and leakage-safe splits are used (Arsa et al., 2022; Sandoval-Pillajo et al., 2025). In addition, our added a reasoning step that lifted both stability and minority-class behavior under difficult optics (e.g., shallow water, shadows). The reasoning particularly helped to achieve high accuracy despite weed detection at the late stage of the crop.

Per-class scores (Table 4) show Rice with near-ceiling performance (Precision = 0.9875; Recall = 0.9820; IoU = 0.9700), while Weed retains strong but lower IoU (0.8199) and balanced precision/recall (~0.90). Others is the most variable (IoU ≈ 0.86), reflecting mixed backgrounds (wet soil, levees/paths, boards/stubble). These per-class IoU values further confirm that the proposed framework improves Weed detection while maintaining high Rice accuracy.

These patterns fit agronomic and sensing expectations at canopy closure: rice forms coherent, row-aligned masses; weeds occur as edge clusters and small inter-row patches; "others" span heterogeneous, sometimes specular or shadowed surfaces that can mimic vegetation unless explicitly gated. The vegetation gate and dark-pixel guard were introduced precisely to handle these confounders, given that water-surface glint and variable illumination are well-known causes of false positives in UAV RGB mapping. Recent studies confirm that variable illumination, background heterogeneity, and water-surface glint are

Table 3
Summary metrics for the WeedGraphNet model (OA, Micro-/Macro-F1, mIoU).

Metric	Value
Overall Accuracy (OA)	0.9742
Micro-F1	0.9742
Macro-F1	0.9368
Mean IoU (mIoU)	0.8833

Table 4
Per-class performance in the WeedGraphNet.

Class	Precision (User's Acc.)	Recall (Producer's Acc.)	F1 (Dice)	IoU
Others	0.9015	0.9491	0.9247	0.8599
Rice	0.9875	0.9820	0.9848	0.9700
Weed	0.9024	0.8997	0.9011	0.8199

major sources of error in UAV-based vegetation mapping, often leading to misclassification between vegetation and non-vegetation classes if not explicitly addressed (Sun et al., 2024; Wang et al., 2023). WeedGraphNet, thus, responds to these challenges.

Table 5 reports an inference-time ablation study quantifying the contribution of each refinement component. Relative to the baseline U-Net (A0), vegetation gating (A1) improves OA from 0.9668 to 0.9699 and increases Weed IoU from 0.7600 to 0.7850. Adding the chromatic prior (A2) further improves Macro-F1 to 0.9280 and Weed IoU to 0.8020, while adding geometric/component constraints (A3) improves Weed IoU to 0.8080. The full WeedGraphNet (A4) achieves the best overall performance (OA = 0.9742, Macro-F1 = 0.9368, mIoU = 0.8833) and the highest Weed precision and IoU (0.9024 and 0.8199), indicating complementary gains from combining the refinement modules.

Table 6 compares WeedGraphNet against a generic morphological cleaning baseline applied to the U-Net output. Morphology cleaning provides only modest improvements over the baseline (mIoU 0.8610 vs 0.8580; Weed IoU 0.7680 vs 0.7600), whereas WeedGraphNet yields substantially higher performance (mIoU = 0.8833; Weed IoU = 0.8199). This demonstrates that the gains are not explained by standard morphological cleanup alone, but by the integration of vegetation gating, chromatic priors, and agronomic structure constraints.

Fig. 5 summarizes the quantitative findings, including contextual comparison, ablation analysis (A0–A4) across metrics, and comparison with generic morphological post-processing. It is noted that comparisons with prior studies are intended for contextual reference only, as differences in datasets, class definitions (e.g., inclusion of non-vegetation classes), and evaluation protocols may influence reported metrics. Therefore, performance improvements are primarily assessed through controlled comparisons within this study, where all models share identical data and class definitions. In this context, per-class IoU values (Table 4) provide a more detailed and reliable assessment of segmentation performance across categories.

4.2. Confusion-matrix analysis

The confusion matrices shown in Fig. 6 provide detailed insight into class-specific performance and error patterns. Both count-based and row-normalized matrices are presented to illustrate absolute misclassification counts and relative class-wise confusion, respectively. The confusion matrices in Fig. 6 reveal two dominant error modes. First, a portion of Weed pixels is misclassified as Rice, particularly in dense canopy regions where spectral differences are reduced. This is reflected by the confusion between Weed and Rice classes (e.g., 33 Weed pixels predicted as Rice, and a small proportion in the normalized matrix). Second, confusion occurs between Others and vegetation classes, primarily due to heterogeneous backgrounds such as water, levees, and shadows that can visually resemble vegetation under RGB imaging. These patterns are consistent with the challenges of late-season conditions, where reduced contrast and environmental artifacts affect class separability.

Overall, the relatively low confusion among other class pairs indicates that the proposed refinement framework effectively improves class separability and suppresses implausible predictions under challenging field conditions.

4.3. Learning dynamics of WeedGraphNet and baseline comparison

Fig. 7 illustrates the learning dynamics of the proposed framework compared with baseline models under different training configurations. Learning-curve comparisons show that all models reach high precision/accuracy rapidly (by ∼ 30–50 epochs / ∼1000–1500 iterations), then plateau. WeedGraphNet stays consistently above three U-Net baselines that differed only in regularization and augmentation strength, indicating that knowledge infusion improves convergence and end-point accuracy without changing backbone capacity. This agrees with prior observations that lightweight post-processing with morphology and context can stabilize semantic maps when appearance cues are weak, and it complements literature showing that row-aware reasoning improves agricultural perception under occlusion and variable lighting (J. Li et al., 2022a; Luo et al., 2023).

4.4. Qualitative mapping results of WeedGraphNet

Qualitative segmentation results are presented in Figs. 8 and 9 to illustrate model behavior under typical late-season conditions. Patch-level examples (Fig. 8) illustrate three typical late-season scenarios:

- Inter-row escapes: thin weed streaks along rice boundaries are preserved (edge-aware rule), reducing false negatives.
- Levees/paths: long black strips remain Others after VI-gate + continuity checks, suppressing vegetation hallucinations. They are located nearby strips of weeds (colored red) or unvegetated areas away from rice plantations (gray).
- Shallow-water glint: bright highlights are masked by the glare prior derived from ground photos, preventing "weed" spikes on mirror-like pixels.

Table 5
Ablation study quantifying the contribution of refinement components (VG = vegetation gating; CP = chromatic prior; GC = geometric/component constraints).

Configuration	VG	CP	GC	OA	Micro-F1	Macro-F1	mIoU	Weed Precision	Weed IoU
A0 Baseline U-Net (appearance only)	No	No	No	0.9668	0.9668	0.9070	0.8580	0.8620	0.7600
A1 + VG	Yes	No	No	0.9699	0.9699	0.9180	0.8690	0.8860	0.7850
A2 + VG + CP	Yes	Yes	No	0.9716	0.9716	0.9280	0.8760	0.8940	0.8020
A3 + VG + GC	Yes	No	Yes	0.9722	0.9722	0.9290	0.8780	0.8960	0.8080
A4 Full WeedGraphNet	Yes	Yes	Yes	**0.9742**	**0.9742**	**0.9368**	**0.8833**	**0.9024**	**0.8199**

Table 6
Comparison against a generic post-processing baseline beyond architectural variants.

Method	OA	Micro-F1	Macro-F1	mIoU	Weed Precision	Weed IoU
Baseline U-Net	0.9668	0.9668	0.9070	0.8580	0.8620	0.7600
U-Net + morphology cleaning	0.9687	0.9687	0.9100	0.8610	0.8720	0.7680
WeedGraphNet (full)	**0.9742**	**0.9742**	**0.9368**	**0.8833**	**0.9024**	**0.8199**

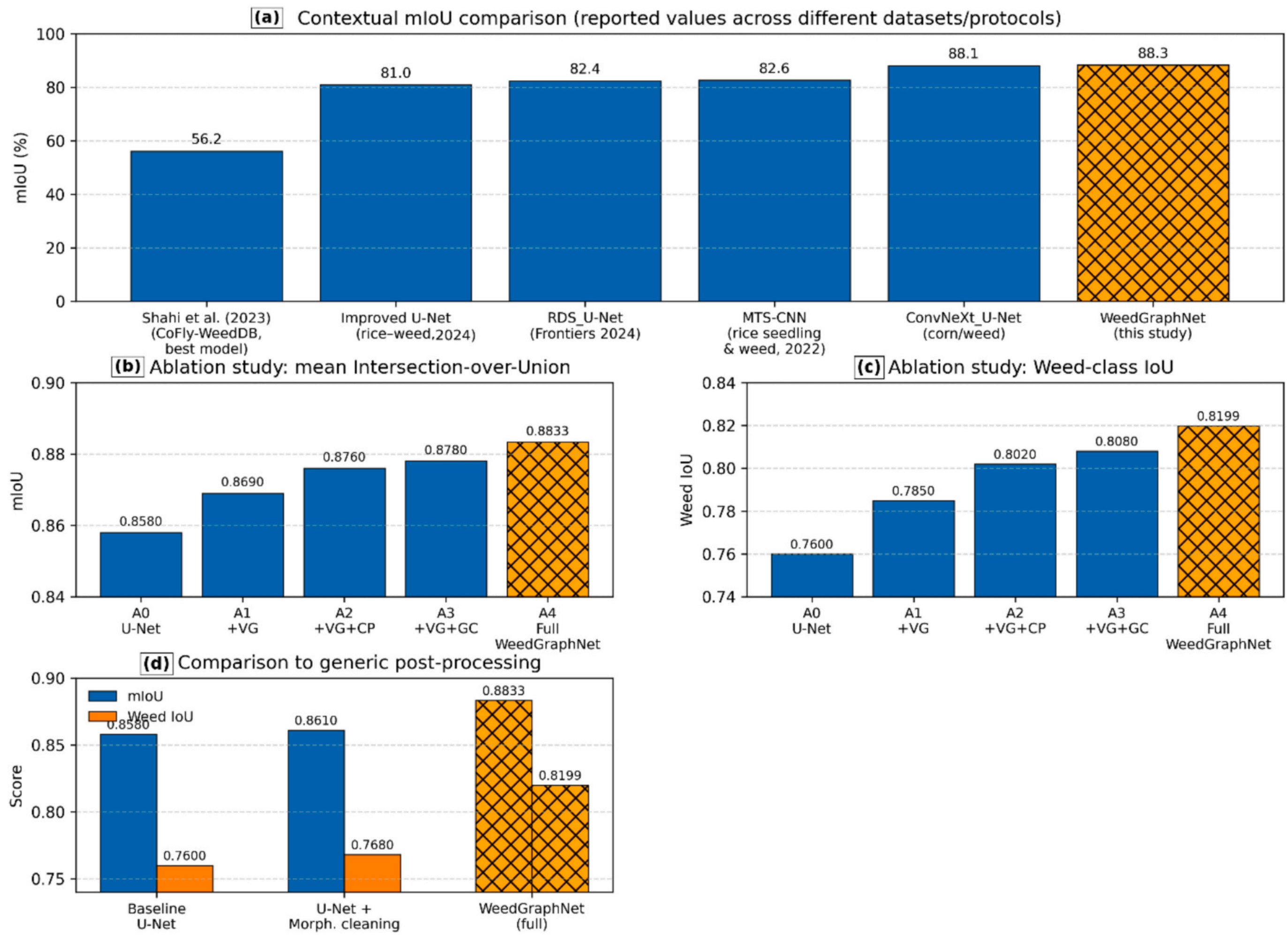

Fig. 5. Contextual comparison, ablation analysis, and baseline post-processing comparison for WeedGraphNet. (a) Contextual mIoU comparison using reported mIoU values from selected UAV crop–weed segmentation studies (Castellano et al., 2023; Celikkan, 2025; Shahi et al., 2023) and the proposed WeedGraphNet (this study). Values are shown as reported in each paper and are not directly comparable because datasets, crops, sensors, class definitions, and evaluation protocols differ. (b) Ablation study (mIoU): incremental contributions of the refinement components from the baseline U-Net (A0) to the full WeedGraphNet (A4). (c) Ablation study (Weed-class IoU): effect of each component on the weed class specifically. (d) Comparison to generic post-processing: baseline U-Net vs. U-Net + standard morphological cleaning vs. full WeedGraphNet, showing that gains are not attributable to generic post-processing alone.

Fig. 9 **presents consistent class representations for both ground truth and predictions, while** Fig. 8 **focuses on illustrative qualitative cases, with visualization adapted to emphasize model behavior. Additionally,** Fig. 9 shows the refinement stage improves late-season robustness by suppressing false positives on non-vegetated artifacts (e.g., water, levees, shadows) while preserving agronomically plausible weed patterns. Weed occurrences are not confined to inter-row spaces but also appear as irregular patches within the canopy and along field boundaries, highlighting the need for spatially explicit mapping. These qualitative behaviors are consistent with field observations and prior reports that late-season weeds often cluster along edges and inter-row gaps, and that RGB-based mapping is sensitive to glint and shadow effects (Guo et al., 2024b; Kawamura et al., 2021; Li et al., 2024a; Wang et al., 2024).

To assess spatial transfer within the study site, the dataset was partitioned using a blocked spatial split, ensuring that training and validation tiles were spatially disjoint (Section 2.6). Consequently, the performance reported in Tables 3 and 4 reflects prediction quality on unseen geographic regions of the orthomosaic rather than adjacent or overlapping image content, providing an initial test of spatial generalization within the field.

Regarding large-area applicability, the proposed workflow is inherently scalable to full UAV orthomosaics. The combination of fixed-size tiling, leakage-safe processing, and georeferenced input–output handling allows the U-Net inference and subsequent knowledge-guided refinement to be applied sequentially across arbitrarily large mosaics. Outputs can be reconstructed as GIS-ready georeferenced raster products, including a three-class thematic map (rice, weed, others) and per-class probability layers, all aligned to the coordinate reference system of the input imagery.

In this study, model training and quantitative evaluation were conducted on a representative, labeled subset of the full orthomosaic due to the substantial annotation effort required for late-season canopy conditions. Nevertheless, the complete georeferenced orthomosaic is available, and no methodological constraints prevent extending inference to the full field or to multiple fields from different locations captured under similar acquisition settings. This design supports operational deployment at the field scale and facilitates integration into geoinformation workflows such as spatial summarization, prescription mapping, and cross-date monitoring.

5. Discussion

This study addresses a recurrent limitation in UAV-based weed

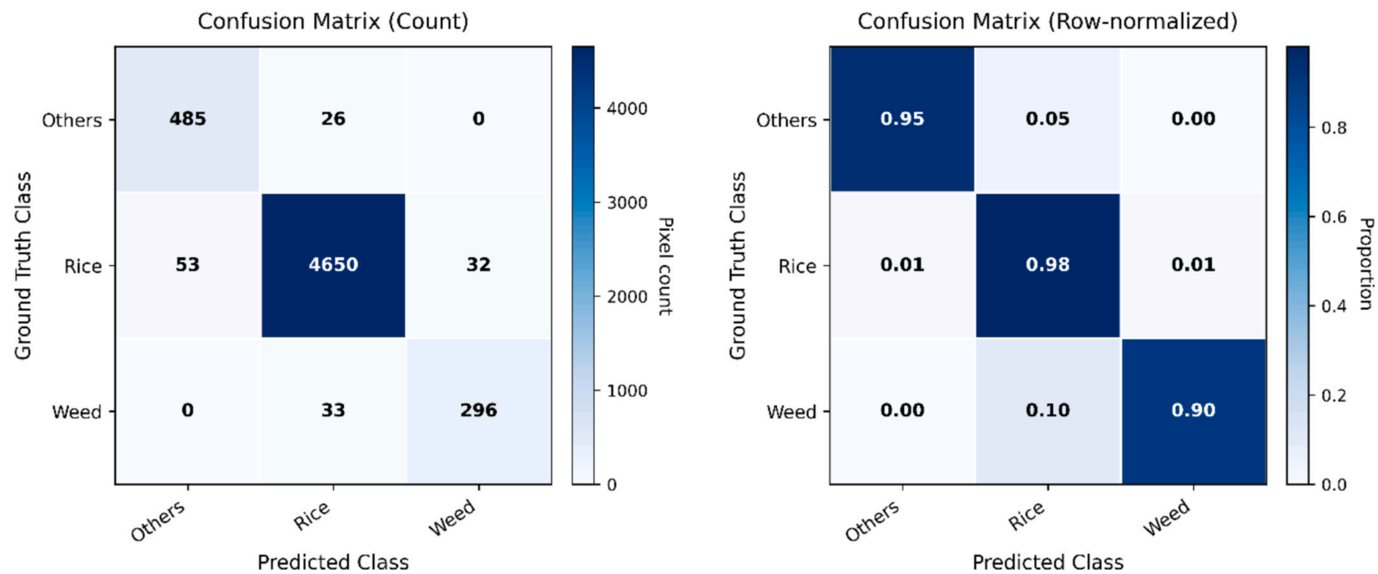

Fig. 6. Confusion matrices derived from the proposed framework outputs (left: count-based; right: row-normalized). Rows correspond to ground truth labels and columns correspond to predicted classes (Weed, Rice, Others).

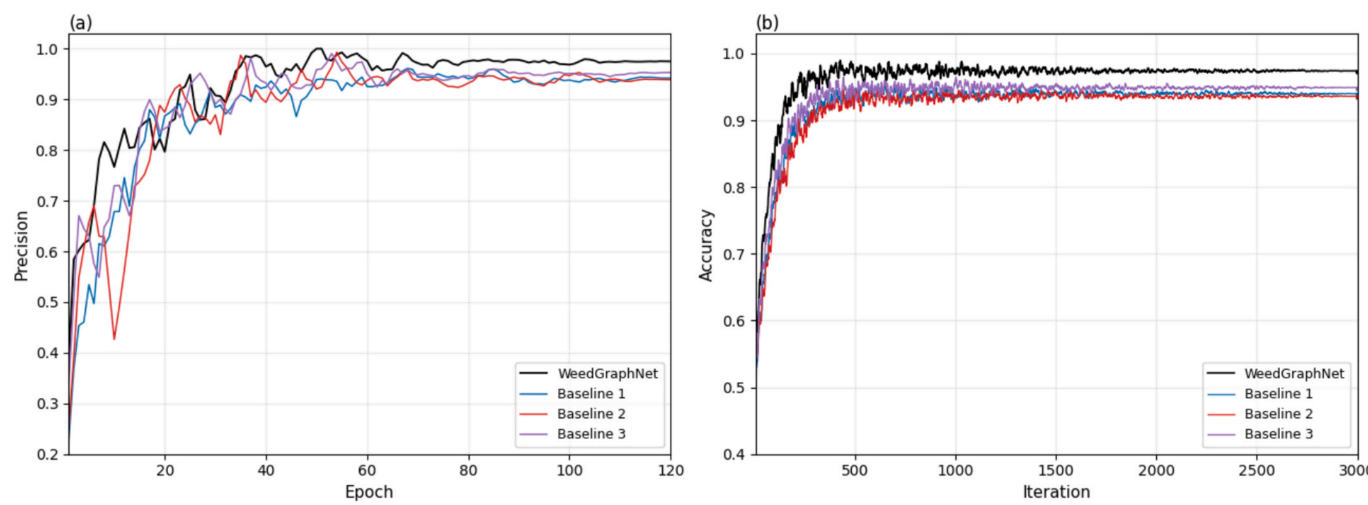

Fig. 7. Learning dynamics of WeedGraphNet compared with three U-Net baselines trained with different regularization and data augmentation settings (light, moderate, and strong). The curves show (a) precision versus epoch and (b) accuracy versus iteration.

mapping: performance degradation under late-season canopy closure, when occlusion, flooded surfaces, and illumination artifacts undermine appearance-driven segmentation. WeedGraphNet combines a compact U-Net with an interpretable refinement stage that encodes acquisition-aware and agronomic priors. The completed ablation study and baseline comparison show that each refinement component contributes measurably, and that the full system outperforms both the appearance-only baseline and a generic morphology-cleaning post-processing baseline. From an Earth Observation and geoinformation standpoint, the method is designed for georeferenced orthomosaics and produces GIS-ready outputs, enabling integration into mapping and decision workflows beyond small-patch demonstrations.

5.1. *Standing within UAV weed mapping literature*

Most UAV-based weed mapping studies report strong performance during early growth stages, when crop–weed separability is high and canopy structure remains open. Under such conditions, convolutional neural networks trained on RGB or multispectral imagery can generate accurate weed maps and support site-specific management and prescription mapping. Numerous studies demonstrate near-ceiling performance in early phenology, particularly when background complexity is limited and illumination conditions are favourable (Mohidem et al., 2021; Qu and Su, 2024).

However, comparative analyses and systematic reviews consistently emphasize that segmentation accuracy degrades as crop development advances and scenes become more complex due to canopy closure, background heterogeneity, and acquisition variability. Late-season conditions are both less frequently evaluated and more prone to misclassification, particularly for minority weed classes embedded within dense crop canopy or near water-covered surfaces(Hunter et al., 2020; Johnson and Mueller, 2021).

By explicitly focusing on late-season rice paddies under canopy closure, this study addresses a documented gap in the UAV weed-mapping literature. Rather than pursuing larger or more complex backbones, WeedGraphNet extends existing approaches by integrating knowledge-guided refinement, enabling stable predictions in conditions where appearance-only models are known to fail.

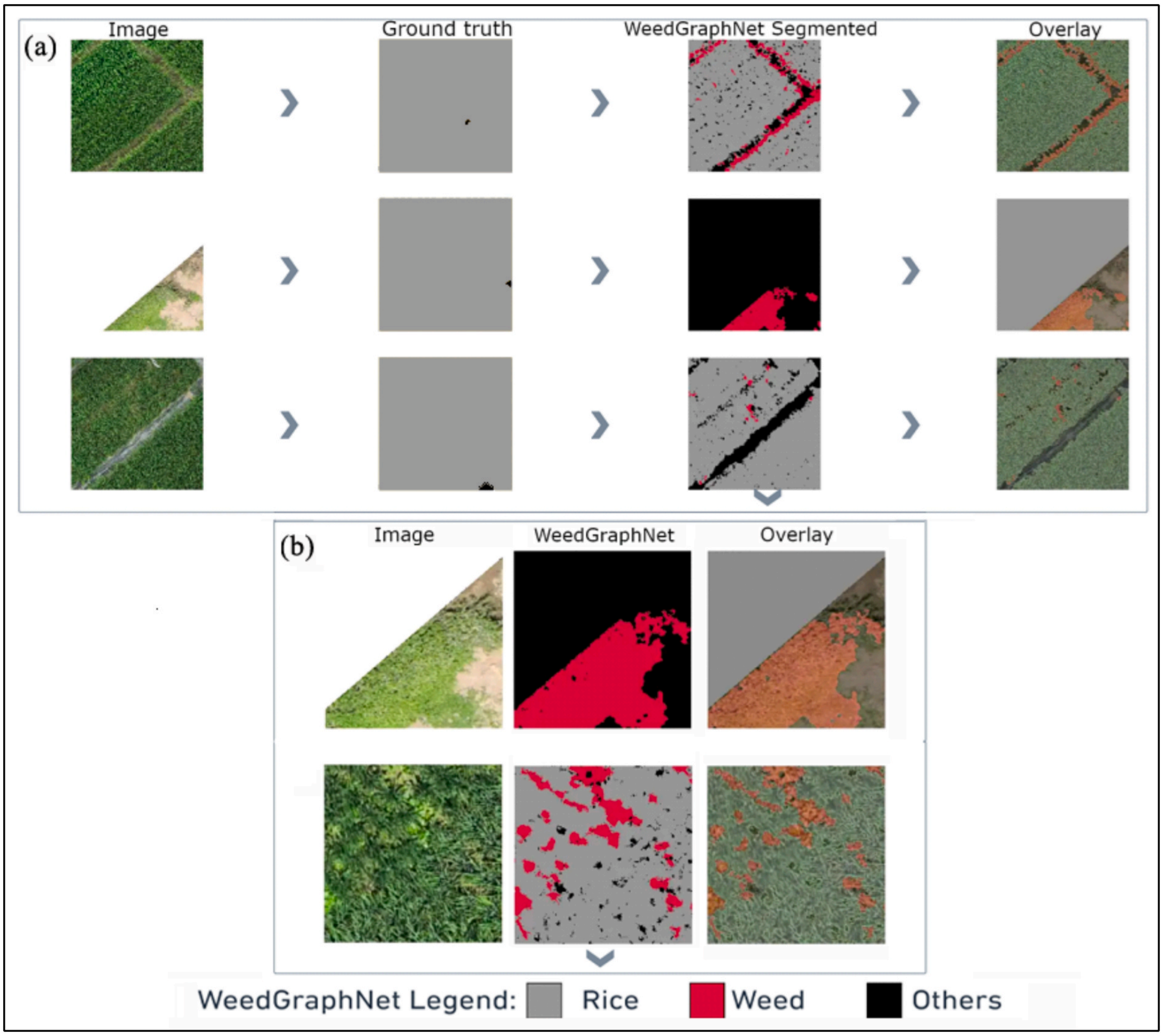

Fig. 8. Qualitative segmentation results. (a) Representative examples showing RGB image, ground truth, prediction, and error overlay highlighting differences between prediction and ground truth. (b) Zoomed examples illustrating inter-row weed detection and suppression of false positives. A consistent color scheme is used across all panels (Rice: gray; Weed: red; Others: black). Note that some regions contain no weed instances in the ground truth, and are included to demonstrate the model's ability to avoid false positives. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

5.2. Knowledge-guided refinement under canopy closure

Late-season rice environments present a combination of confounding factors that are difficult to resolve using RGB appearance alone: (i) spectral mixing and occlusion within dense canopy, (ii) shallow water and wet soil that introduce specular reflections, and (iii) spatially variable illumination and shadows across orthomosaics. These effects are widely reported as major error sources in UAV and airborne vegetation mapping, often leading to false vegetation detections on water, levees, and shaded surfaces (Mishra et al., 2023; Ortega-Terol et al., 2017).

Recent studies emphasize that explicitly modeling acquisition conditions such as illumination variability, glare, and shadowing can substantially improve segmentation stability, particularly when only visible bands are available (Cho et al., 2025; L. Guo et al., 2024a). The refinement stage in WeedGraphNet directly responds to these findings by combining vegetation gating with a dark-pixel safeguard, soft chromatic priors, and component-level plausibility constraints. These mechanisms suppress implausible vegetation responses while preserving weak but agronomically meaningful weed signals, explaining the reduction in false positives over non-vegetated surfaces and dense crop interiors.

5.3. Agronomic structure, Earth observation perspective, and scalability

Agronomic structure provides a complementary signal when pixel-level separability collapses. Crop row geometry is spatially coherent and persistent, while weed escapes at late growth stages often occur between rows, along margins, and near irrigation structures. Prior work in both weed detection and agricultural robotics demonstrates that row-aware reasoning can significantly improve robustness under occlusion and background clutter (Fan et al., 2023; X. Li et al., 2022b; Zhang et al., 2022).

From an Earth Observation and geoinformation perspective, WeedGraphNet is explicitly designed around georeferenced UAV orthomosaics and produces GIS-ready raster outputs, including class maps and per-class probability layers. Such outputs are essential for downstream spatial analysis, prescription mapping, and temporal monitoring, which distinguishes operational EO workflows from small-patch computer vision demonstrations. Tile-based inference with bounded memory usage further enables processing of large orthomosaics, aligning with established best practices for scalable UAV and airborne remote sensing applications (Belgiu and Drăguţ, 2016; Maxwell et al., 2018).

The combination of high Rice IoU (0.97) and Weed IoU (0.82)

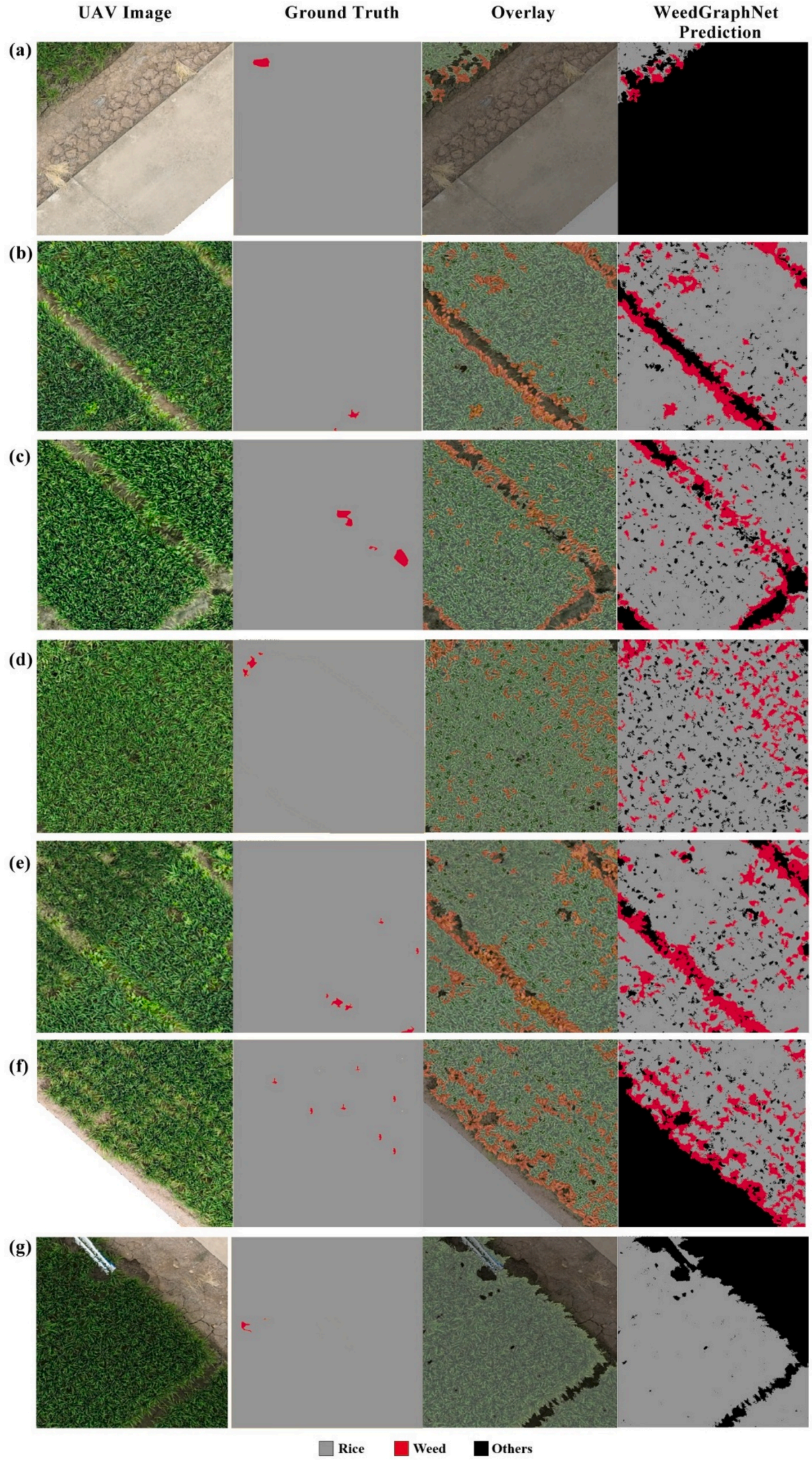

Fig. 9. Sample regions (a–g) showing inter-row weed patterns under late-season canopy closure. Ground truth and prediction maps use a consistent color scheme (Rice: gray; Weed: red; Others: black), ensuring clear interpretation of class distributions.Transferability and large-area applicability. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

suggests reliable delineation of spray-eligible patches without over-treating the crop interior key for reducing chemical footprints while targeting residual escapes. Literature on deep learning weed detection notes that RGB-only pipelines are attractive for smallholder contexts but are particularly sensitive to lighting and background artifacts (Murad et al., 2023); the photo-prior and reasoning design directly addresses these weaknesses while keeping the system interpretable and sensor-light (Gautam et al., 2025).

The proposed framework has practical implications for site-specific weed management. By accurately delineating weed-affected areas, it enables targeted herbicide application and reduces unnecessary chemical use. This is particularly relevant under late-season conditions, where weeds are spatially heterogeneous and often intermixed with non-target surfaces such as water, soil, and levees.

5.4. Transferability, extensibility, and limitations

Transferability encompasses robustness to spatially new areas within a field and generalization across fields, dates, sensors, and acquisition conditions. In this study, a blocked spatial split ensures that training and validation tiles are spatially disjoint (Section 2.6), providing an initial test of spatial generalization within the orthomosaic. Consequently, the performance reported in Tables 3 and 4 reflects prediction quality on unseen geographic regions rather than adjacent or overlapping image content.

More broadly, the refinement module relies on generalizable cues, including vegetation separation, illumination and glint handling, and structural plausibility, which have been shown to remain effective across diverse agricultural landscapes(Weiss et al., 2020).

he framework is also modular and extensible. While this study focuses on an RGB-only operational setting, multispectral indices (e.g., NDVI or red-edge) could further strengthen vegetation gating when available. In addition, UAV-derived structural products (DSM/CHM) can assist in separating low weeds from crop canopy, and multi-temporal UAV or satellite observations can introduce phenological constraints that improve robustness under appearance ambiguity. Such extensions are consistent with broader EO trends toward multi-sensor and multi-temporal data fusion in agricultural monitoring (Wang et al., 2024).

Regarding large-area applicability, the proposed workflow is inherently scalable to full UAV orthomosaics. The combination of fixed-size tiling, leakage-safe processing, and georeferenced input–output handling allows the U-Net inference and subsequent knowledge-guided refinement to be applied sequentially across arbitrarily large mosaics. Outputs can be reconstructed as GIS-ready raster products, including a three-class thematic map (rice, weed, others) and per-class probability layers aligned to the coordinate reference system of the input imagery.

Limitations include reliance on a single acquisition campaign and a limited labeled region, as well as the heterogeneous nature of the "Others" class. Future work should extend evaluation to multi-field and multi-date datasets and explore finer semantic subdivision of non-vegetated classes to further reduce confusion under complex background and illumination conditions. Despite these limitations, the availability of the full georeferenced orthomosaic and the modular design of the framework support extension to larger areas and integration into operational geo-information workflows such as spatial summarization, prescription mapping, and cross-date monitoring.

6. Conclusion

This study introduced WeedGraphNet, a knowledge-infused segmentation framework for late-season rice–weed mapping under canopy closure using high-resolution RGB UAV imagery. By coupling a compact U-Net with a lightweight, interpretable refinement stage incorporating vegetation gating, soft chromatic cues, edge- and row-aware context, and morphological constraints, the framework achieved robust performance on held-out tiles (OA = 0.974, Macro-F1 = 0.937, mIoU = 0.883). Per-class results (IoU = 0.97 for rice and 0.82 for weeds) demonstrate improved stability and reduced false positives relative to appearance-only baselines, particularly over shallow water, levees, and dense crop interiors.

The main contributions of this work are threefold:

1. A hybrid, reasoning-based refinement strategy that augments deep learning segmentation with interpretable agronomic and photographic priors, improving robustness under late-season imaging conditions where appearance cues are degraded.
2. A sensor-light (RGB-only), georeferenced workflow that produces scalable, GIS-ready outputs suitable for large-area mapping and integration with site-specific and UAV-based weed management systems.
3. Improved delineation of spray-eligible weed patches with reduced overestimation within crop interiors, supporting tighter application footprints and the potential for reduced chemical inputs.

While the current evaluation is limited to a single late-season field site, the framework is modular and extensible. Future work will focus on multi-site and multi-temporal validation, integration of multispectral indices and UAV-derived structural products, and direct quantification of operational benefits such as treated area and herbicide savings. Overall, WeedGraphNet demonstrates that carefully designed, domain-informed reasoning can close key failure modes of late-season UAV weed mapping while preserving interpretability and operational applicability in agricultural Earth observation. By operating directly on georeferenced UAV orthomosaics and exporting GIS-ready probability layers, the framework supports scalable Earth observation workflows for field-to-regional weed monitoring.

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CRediT authorship contribution statement

Muhammad Nasar Ahmad: Writing – original draft, Visualization, Validation, Software, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Xiaohui Yuan:** Writing – review & editing, Supervision. **Shuai Yang:** Visualization, Data curation. **Lichuan Gu:** Supervision. **Xinyu Miao:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jag.2026.105361.

Data availability

The datasets and code supporting the findings of this study are provided as a reproducibility package, including: (i) 512×512 analysis tiles and corresponding label masks, together with the train/validation manifests, (ii) scripts for preprocessing, knowledge-guided refinement, evaluation, and georeferenced export, and (iii) trained model weights and configuration files. If the submission system does not allow uploading the full package, the package will be deposited in a public repository (https://github.com/mnasarahmad/WeedGraphaNet) and cited in the final version of the manuscript. Any raw data not shared publicly (e.g., full-farm orthomosaics or ground photographs that may contain identifiable features of private farmland) can be made available to qualified researchers upon reasonable request.

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